Lesson Notes

MODULE 4 | LESSON 4

Behavioral Types: Behavioral Explanations for Financial Bubbles and Crashes Lesson Notes

Reading Time	90 minutes
Prior Knowledge	Module 4 Lesson 1, Lesson 2, and Lesson 3
Keywords	Barnewall's behavioral model, BB&K behavioral model, Pompian model, momentum, financial bubbles and crashes

In Lesson 3, we learned that behavioral portfolio theory requires that investors' behavioral biases be incorporated in the creation of portfolios. To this purpose, we identified the need to detect whether an individual's biases are mainly cognitive or emotional.

In this lesson, we will learn how to classify investors based on certain behavioral models, which are supposed to produce a better picture of our investors in terms of their ability to bear risk and how to improve the relationship between client and financial advisor/portfolio manager.

In the second part of the lesson, we will look at certain events like financial bubbles and crashes, which should not even exist according to Traditional Finance, and we will explain them from a behaviorist's point of view.

1. Classification of Investors

We have already stressed with sufficient emphasis in the third lesson of this module that, when the behavioral biases of investors are taken into proper consideration, the result is a favorable outcome for the investors.

In that lesson, we also emphasized the shift from traditional finance to behavioral portfolio theory. In TF, clients are *simply unemotional* and equipped with formulaic measures of return and risk. For instance, we could tell a customer that they are better off with a specific portfolio out of several available simply because that portfolio has a higher Sharpe ratio. However, if the client is an individual who reacts emotionally when discussing investments, it is quite likely that they will not find a metric like the Sharpe ratio of much relevance. However, it remains very important to enlist the cooperation of clients as, in the end, they are the only ones who can decide whether to follow our advice or go with their emotional feelings.

Sometimes, a bit of bending of strict TF principles produces much better overall results as clients feel engaged, become easier to advise, and are ultimately happier with their portfolios and more likely to follow strategies and avoid costly mistakes.

We have already discussed the concepts of adapting or moderating in Lesson 3, we also learned to think about the clients' goals and their risk of standard of living, although, mostly in the quizzes, we saw the need to interview customers to ascertain their specific behavioral types.

Behavioral finance has developed a few behavioral models to help identify customers.

1.1 Barnewall's Behavioral Model: Passive and Active Investors

This is a two-way model that dates back to 1987 (see the References section) and in which investors are classified either as passive or active.

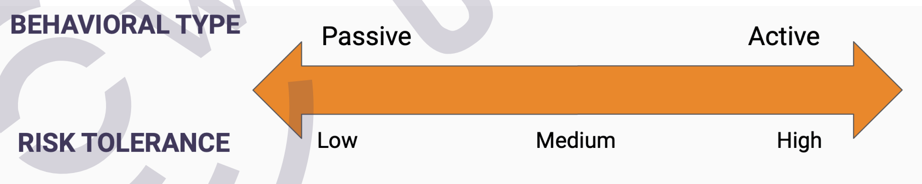
A **passive investor** can be described as an investor who has accumulated wealth through a long and steady employment, very disciplined saving, or by inheriting money.

An **active investor**, on the other hand, is someone who has accumulated wealth by taking an active role in investing their money.

Why is this simple classification important? It has to do with their attitude towards risk:

- Passive investors are more risk averse and are looking for security. They are susceptible to overestimating the risk of investments.
- Active investors are willing to trade some of that security in exchange for higher gains. On the other hand, they are more prone to a behavior that displays overconfidence, which can lead them to underestimate risk. However, whenever they detect a loss of control, their risk tolerance declines rapidly.

Figure 1: Barnewall's Model



1.2 Bailard, Biehl, and Kaiser Model (BB&K)

This behavioral model is a five-way model, introduced in 1986, and is more complex than Barnewall's model. In this model, investor personalities are classified along two axes: **level of confidence and method of action**.

- **Level of confidence** is meant to categorize how much confidence an individual displays when making decisions. An individual with low confidence is labeled as anxious. At the opposite of the spectrum, individuals with high confidence level are defined as confident. There are no intermediate levels, but a financial advisor could notice that a specific individual may fall in between the two extremes.
- **Method of action** is intended to be used to identify whether a specific individual is methodical, analytical, or careful in the way they make decisions or whether they tend to be more on the spontaneous side. An individual who is very methodical and analytical is considered to be careful.

At the opposite end of the spectrum, an individual who is mostly spontaneous is defined as impetuous.

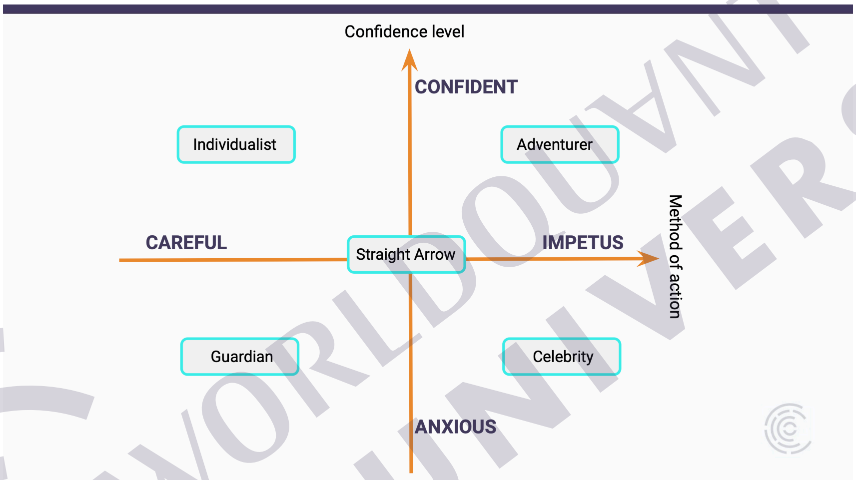
The BB&K model, based on these two dimensions, identifies five different personalities of investors (hence the five-way model):

- Straight arrow;
- Celebrity;
- Adventurer;
- Individualist;
- Guardian.

The fancy names are just that; they are supposed to help associate an image with a client's behavioral type.

At this point, a plot and a synopsis of the model (due to Kaiser) are very useful:

Figure 2: BB&K Model and Its 5 dimensions.



Straight arrow investors can be defined as sensible and secure as they fall in the center of the graph. They are the most balanced of the investor types and are willing to trade some risk in exchange for a suitable amount of additional return.

Guardian investors are careful and anxious. It is a type that is most often associated with individuals who are approaching retirement. Their number one concern is to protect their assets since they have less time to recover from bad investments.

Individualist investors are confident and tend to act in an independent manner. They prefer to make their own decisions after careful analysis and tend to process information rationally.

Celebrity investors have low confidence: They are anxious in fact and tend to be impetuous in the way they act. In practice, this means that we are in the presence of individuals who are aware of their limitations but who act in a spontaneous manner and enjoy being the center of attention. They are not known for being patient.

Adventurer investors are confident and impetuous. They are very difficult to deal with because they are very confident and do not shy away from taking chances. It is not unusual that adventurer investors are usually persons with an entrepreneurial past and who were successful in their business.

What does all this mean for a financial advisor?

Straight arrow investors, guardian investors, and individualist investors are amenable to advice because of their careful behavioral component. However they take advice differently. The guardian type is very willing to listen to advice, especially if they recognize that the advisor has more experience. Individualist investors tend to act on their own when it is time to make decisions.

Celebrity investors are usually seeking out advice, but they may have strong opinions that need to be educated and possibly moderated or corrected.

Adventurer investors are the most difficult of them all to deal with. Highly confident and more than willing to take chances, they are reluctant to take advice.

1.3 Pompian Model

This model was developed in 2008 and it is a four-way model in the sense that it has four behavioral investor types (BITs) at its core: the **passive preserver**, the **friendly follower**, the **independent individualist**, and the **active accumulator**.

The four BITs are identified as such after an interview with clients and can be described as follows:

A **passive preserver** is an individual who places most emphasis on financial security and displays a conservative approach to investing. This investor is generally satisfied with the status quo and believes that preserving one's wealth is preferable to potentially higher returns if those come at a higher risk of losing money. In general, these investors are not prone to either frequent portfolio rebalancing or to excessive trading in and out of positions. Their actions tend to be driven more by how they feel and think, hence they are characterized more by emotional biases rather than cognitive ones. Preservers display mental accounting and accumulate wealth by investing carefully and in a balanced manner.

As preservers have high aversion to risk, they can panic easily and sell out of positions after suffering losses, which often causes them to miss the rebound. Furthermore, their low-risk and cautious approach to investments can produce portfolios that are heavily invested in bonds and cash instruments. Unfortunately, this could be detrimental to them achieving their financial goals.

A **friendly follower** is an individual with low to moderate risk tolerance and who does not have a keen interest in the financial markets. Therefore, these individuals tend to follow the advice of experts or peers when it comes to making investment choices and their decisions are driven by cognitive biases rather than emotional ones. One of their characteristic traits is that they are attracted by the latest investment trends for fear of missing out on something good and, as a consequence, exhibit a faulty understanding of the correct risk they are assuming and the magnitude of potential losses.

They are not likely to indulge in excessive trading in and out of positions or change their portfolio compositions often. The major danger they face is a consequence of their tendency to chase the newest investment advice or trends when they see or hear their peers doing it. As they are followers, they are likely to get into these investments in later stages as they fear missing out on the opportunities and do not really pay attention to whether the timing is right or not. In general, these individuals can be educated on the benefits of portfolio diversification and the advantages of sticking to a long-term financial plan.

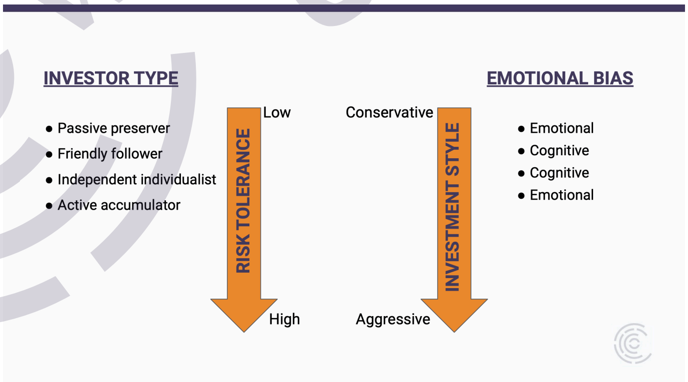
An **independent individualist** is a suitable description for a strong-willed, analytical, and critical thinker who also entertains unconventional ideas about investing. This group of investors is actively engaged in the investment process: They are usually not afraid to do their own research and understand the risks of their investments. They are willing to tolerate risk since they are confident about their ability to research the pros and cons effectively and are not afraid to put in the hours to research their investments.

Overall, we could refer to them as thinkers and doers as opposed to dreamers and followers. However, as they tend to act independently, they are susceptible to confirmation bias in the sense that they tend to ignore certain angles of the problem that run contrary to their views. They also are prone to falling in love with their ideas and are not open to entertaining new ideas and views. Because of this, they would benefit from discussions with their financial advisors as a form of check and possibly a final revision of some of their investment plans.

An **active accumulator** is a description reserved for individuals who make wealth accumulation a priority and are highly engaged in the process. Overall, these individuals possess a strong will and are both decisive and confident in making and following their investment choices, respectively.

These individuals can be recognized for having an entrepreneurial mindset. Often, they have already been successful in their field and are aware of the research and hard work necessary to succeed and apply that approach to investing as well. Nevertheless, their high confidence lvel also causes them to be susceptible to emotional biases like overconfidence and self-control as they dive into excessive trading and portfolio rotations. They seem to enjoy the thrills associated with making good investments and, as a consequence, are attracted to high-risk investments and do not pay enough attention to either asset allocation or diversification.

Figure 3: Pompian Model and Its 4 BITs



Regarding the main types of behavioral biases within each BIT, the most relevant are:

- **passive-preserver**: endowment, loss aversion, status quo, regret aversion, mental accounting, anchoring.
- **friendly follower**: regret aversion, availability, hindsight, framing.
- **independent individualist**: overconfidence, self-attribution, conservatism, confirmation, representativeness, availability.
- **active accumulator**: overconfidence, self-control, illusion of control

Are behavioral types the last word for BPT? The answer is that while they do help, they are not without limitations. In fact, it is possible that one investor displays the traits of more than one type, which makes it difficult to assign them to a specific group. Furthermore, it is not uncommon for individuals to display both emotional and cognitive biases at the same time. And, if that were not enough, the age of investors also plays a role. It is quite admissible that, as investors age, whether they realize it or not, their BIT changes. A portfolio manager or financial advisor should also be careful not to use the models above as if they were strict mathematical models: Two individuals could be classified in the same group, but they may still need different types of advising.

2. Momentum, Fat Tails, Bubbles, and Crashes: The Behaviorists' Explanations

Traditional finance believes that arbitrage has the power to quickly erase any unjustified mispricing —so then why do we have bubbles?

BF suggests that individual investors display an eagerness to imitate the decision of others, a situation called **herding**. However, just because many do it, the fact that many investors exhibit the same behavior does not imply that said behavior is rational.

This behavior can lead to *bubbles* and *crashes* in the most extreme cases. In milder cases, it could explain why stock returns have fat tails. This behavior has been observed more often in the case of illiquid assets where trading is less frequent and investors believe that changes in prices may be a proxy for someone having access to non-public information instead of being caused by lack of liquidity.

Momentum arises when price movements are correlated with their past history. Several studies have highlighted the fact that over shorter periods (1–2 years), there is positive correlation, which, however, tends to disappear and even become negative over longer periods (2–5 years) as prices revert to their means.

Momentum has been invoked to partly explain short-term underreaction to relevant information and longer term overreaction.

The behavioral bias that causes it is anchoring. In fact, investors attach a special significance to the price at which they bought the asset. When they believe in mean reversion, they are initially underreacting to new information and choose to believe that the new higher price has actually introduced new higher risk as well. Hence, it takes time for stock prices to adjust to new positive information.

According to behaviorists, the anchoring bias is further reinforced by availability bias, which in this particular instance could be labeled as a **recency effect**. When investors see that a stock goes up for several days in a row or even over a period of weeks, they extrapolate that the stock will continue to go up in the future as well and thus do not feel a particular sense of urgency to buy as they are essentially seeing a trend that may or may not be there.

After a period of time where investors have been on the sidelines, they start to suffer from regret aversion, the feeling that they might be missing an opportunity or that everyone else is taking advantage of a great opportunity and they are among the few who have not done so yet. During this period, there is also some hindsight bias as investors who have not bought the asset blame themselves for not having noticed the development and having failed to realize what was a very easy-to-spot trend. At this point, feeling regret and blaming themselves for missing the trend, they rush to remedy the issue. This action creates a **trend-chasing effect** where investors rush to buy assets they wish they had owned already.

At some point, evidence collected over the years has shown that over longer periods of a few years (3–5), mean reversions take place.

When corrections begin, investors experience loss aversion and therefore hold on to stocks that are losing value. This behavior is also called a **disposition effect**. Psychologically, they hope that the loss in value is just a temporary effect and that soon assets will start to go up again, and this attitude slows down the correction process. We can recognize some anchoring bias as well while investors try to rationalize the losses. Players like hedge funds could even be trying to sell short to capitalize on what is perceived as a temporary correction at best.

Eventually, particularly in the case of bubbles, as the bubble unwinds, investors begin to realize what is going on and capitulate. Now, fearing to be among the last ones still holding a bad asset, they rush to sell and accelerate the price decline. In particular, since we mentioned hedge funds, when the average investors see hedge funds selling short assets, they are susceptible to think that hedge funds managers must know something they don't and that points to prices falling further. Thus, this particular sort of negative momentum can also trigger a rout.

Note: TF has tried to explain momentum without using behavioral biases. Unfortunately, the bibliography is quite extensive, and this is not the place to get into it. We mention that there is also an extension of the Fama French three-factor model, called the **Fama French and Carhart model,** in which momentum is added as an additional factor to the model.

3. A Working Example

Thunder & Bolts & Associates (TBA) is an investment management firm that has been in operation since 1985 and uses proprietary valuation models based on fundamental analysis to select individual stocks and bonds for the actively managed portfolios of their clients. During the last 10 years, TBA has been taking into account investors' behaviors and past experiences when advising its clients.

Muinul Chowdhury is a young professional who recently earned a M.Sc. degree in Finance from WQU and has since been hired by TBA in their portfolio management team. He has been assigned three new clients whose portfolios need to be transitioned to TBA's managed accounts. To this purpose, Mr. Chowdhury is reviewing the completed questionnaire for each of them and prepares three short bios:

Colby Skelton, 40 years old, earns his income from royalties from his successful travel books. He has accumulated a portfolio with a current value of $3.0 million. Up to now, Mr. Skelton has managed his own portfolio and feels very confident about his own ability to invest successfully when good opportunities present themselves. Still, in several instances, Mr. Skelton's portfolio contained positions with sizable losses that he proved reluctant to sell. As a matter of fact, Mr. Skelton stated that, due to the size of the losses and the size of his portfolio, he decided to seek professional help. He acknowledges that he would be agreeable to considering high-risk investments as long as his research indicates that these investments represent an attractive opportunity. Currently, his portfolio consists of 12 equity positions of different dollar values, and these equities are diversified across seven different industries and 4 different countries.

Jayne Griffith is 60 years old and has steadily risen through the ranks of a government agency after working there for nearly 30 years, and she is a now a deputy director. She currently has a financial portfolio valued at $1.2 million. Mrs. Griffith has little experience in investing and has come to TBA as she is nearing retirement, and about 75% of her portfolio is invested in a single company that her uncle started and manages. She is specifically worried that an economic downturn could affect the company and her retirement plans. She stated also that, at this stage of her life, she would prefer to avoid any high-risk investments.

Neal Johnson is 50 years old and has been very successful starting his own high tech company when he was only 22. He found a very small niche for his company with little competition that he has exploited very well. As a result, Mr. Johnson earns several million dollars each year. So far, Mr. Johnson has managed his own portfolio via an online brokerage account. He invested about $20 million in it and his portfolio currently has a value of $19 million. He admits to having limited investment expertise and, despite his high-tech business, he considers himself a limited to moderate risk taker when it comes to investing his money. He explains that he has been selecting his investments mainly by acting on the advice of other managers in his field, from what he learned by attending sector conferences, and by taking advantage of opportunities he hears about at the many social events he attends. Mr. Johnson has purchased almost exclusively the most popular stocks for his portfolio and is invested completely in shares of 10 high-tech U.S. companies. Mr. Johnson complains that, due to his very busy schedule, he has often missed the right time to buy and sell stocks or, otherwise, he would have "made a killing" in the stock market.

Question: Which of the 3 clients is most likely a friendly follower?

Neal Johnson. This can be determined from the following elements:

- He stated that he has a low to moderate risk tolerance;
- he follows the advice of other high-tech professionals, friends, and acquaintances and possibly financial advisors; and
- the majority of his investments are in popular assets without checking whether these are truly suitable for his long-term goals.

Home bias is defined as a bias where an investor prefers investing in their domestic country vs. foreign countries or prefers to invest in companies with which they are familiar over companies that they do not know. **Familiarity investment** (bias), instead, is defined as the preference to stay in one's comfort zone, i.e., for instance investing in a sector in which one has invested before. **Naive diversification** is when an investor allocates approximately the same amount to each asset class. Finally, another behavioral bias that is often mentioned in the field is **gambler's fallacy,** which can be described as a situation in which an individual believes that a random event is less likely to happen following an event or series of events.

Question: Considering the following biases: home bias, loss aversion, naive diversification, and familiarity investment, what is the most likely behavioral bias for Mr. Skelton, judging from his portfolio? And what would you say the least likely bias is?

The most likely bias is loss aversion. The least likely is home bias. The reasoning is that:

- Mr. Skelton has displayed reluctance to sell positions that have lost value, which is evidence of loss aversion.
- Mr. Skelton's portfolio contains equities of four different countries, hence he is not displaying home bias.
- Mr. Skelton is willing to consider new investments, even high risk investments so there does not seem to be a case for familiarity investment. But this is less clear from the information we have available.
- Mr. Skelton does not display naive diversification as his allocations have different dollar values. As such, it is difficult to agree on his attitude towards naive diversification.

Patrick Hess, Ph.D., CFA is a senior portfolio manager at TBA and has had a long successful performance record. He has voiced his concern about using BIT models as they have many limitations. He is also supervising Mr. Chowdhury and explains to him that, although TBA uses fundamental research and it provides great value to its customers, technical analysis is also compatible with sound investment advice. Upon hearing this statement, another colleague, Devin Wirawan, CFA, feels compelled to jump into the discussion and states that TBA fundamental value models incorporate hundreds of pieces of information, and as such, they are more sophisticated and thorough than any other analysis. Finally, he says that he believes that the opportunities identified with technical analysis are actually not real market anomalies, but rather the result of under-estimating true risk exposure. Dr. Hess counters by saying that technical analysis has the potential to detect opportunities in cases where markets are over or underreacting to new relevant information. Mr. Chowdhury then states that from what he has seen in his still relatively short career, every time a sector falls below its 10-year average level, the sector has recovered to its mean in a relatively short period of time.

Question: Concerning Dr. Hess' statement regarding behavioral types, it is most likely justified because individual investors

1. exhibit characteristics of several investor types;
2. retain the same behavioral biases as they age;
3. display primarily emotional or cognitive biases, but not both.

The answer is 1; 2 and 3 are clearly wrong.

Question: Based on Mr. Wirawan's tendency to strictly adhere to TBA's proprietary valuation models, he is most likely to exhibit overconfidence in TBA's models because of:

1. availability bias
2. self-attribution bias
3. an illusion of control

The most likely explanation is 3. Mr. Wirawan's belief that, through the model he uses, he is in a position to use new information better than his competitors. The fact that the models use a lot of data could actually reinforce his illusion of control. While, in general, more information is better than less information, the use of complex models does not necessarily say a lot about how accurate the forecasts have been in the past. There is really no indication of availability bias and Mr. Wirawan is not self-attributing the success of TBA's models.

Question: Regarding technical analysis, which of the three individuals has made the most accurate statement?

Given the three statements, Mr. Chowdhury made the most accurate statement. What he described is momentum investment and there is a certain amount of evidence that markets display short-term underreaction and long-term overreaction to information.

4. Seven Sins of Fund Management

Individual investors, analysts, and portfolio managers have one thing in common: They are all first and foremost individuals and, as such, they display behavioral biases. When we invest, we do not need to prove our ideas in a mathematical sense, and therefore, we tend to indulge in faulty reasoning mixed up with a lot of feelings.

While BF has been around over 40 years, it was not widely known among practitioners outside of academia. That began to change in the late 1990s and even more so in the early 2000s.

In 2005, James Montier an economist at Dresdner Kleinwort Wasserstein in London—later co-Head of Global Strategy at Société Générale, and, more recently, member of GMO's asset allocation team, Montier has often been voted a top-rated strategist in the annual Thomson Extel survey—published a research note, **Seven Sins of Fund Management: A Behavioral Critique**, which flew off the shelves: It had to be reprinted many times over a period of several months.

In a somewhat iconoclastic style, Mr. Montier provided a very effective way to analyze and think of the most egregious behavioral traits that often impede most individuals (day traders, portfolio managers, financial analysts, etc.) from being truly good at investing. The table that follows is an abridged version of Montier's Research Report from 2005.

"SIN"	DESCRIPTION
Forecasting (Pride)	There is a large body of evidence that people are simply not good at forecasting. However, about 75% of fund managers think that they are above average at their jobs! The inability to forecast seems to be rooted in the fact that we all seem to be over-optimistic and over-confident.
Illusion of Knowledge (Gluttony)	Like the robot Johnny 5 in the 1986 movie *Short Circuit*, individuals thirst for more and more inputs/information. Investors display the belief that they need to know more than everyone else in order to outperform. This is probably a consequence of having been taught from an EMH point of view. Unfortunately, the verdict from the field seems to be that increasing the amount of information gathered serves mainly to increase our confidence rather than increase our accuracy.
Meeting Companies (Lust)	Analysts meet with companies' management. Individual investors read their research reports or companies' reports. There are at least five hurdles that must be overcome if meeting companies is to add value to an investment process: 1. More information is not necessarily better information. 2. Corporate managers are not perfect and have their own behavioral issues, often in the form of cognitive illusions. 3. We suffer from confirmation bias and tend to look mostly at the information that agrees with us and discard the rest. 4. We naturally tend to obey figures of authority. Top management, being at the top, inspires such feelings. Hence, analysts and fund managers are effectively over-awed and take their words for it. 5. We are lousy at discerning truth from deception, although we tend to believe that we can spot liars.
Thinking you can outsmart everyone (Envy)	Over-optimism and overconfidence are likely the main culprit if we are prone to thinking along the lines of "Now I understand Behavioral Finance; I can outsmart everyone else." Plenty of experiments have shown that we are much worse than we think.
Short time horizon and over-trading (Avarice)	Too many investors confuse noise with news and try to outsmart each other and end up with time horizons that are too short (usually 11 months for stocks on NYSE at the time of the research). Over such a short time span, returns have nothing to do with intrinsic value or discounted cash flows and are just a function of price changes.
Believing everything you read (Sloth)	Stock brokers spin stories, which act like sirens drawing investors onto the rocks. We all love a story and we appear to be hardwired to accept stories at face value, even some very improbable ones.
Group based decisions (Wrath)	We tend to accept a principle according to which groups are better at making decisions than individuals. Ideally, groups meet to exchange ideas and reach sensible conclusions. Yet, research by psychologists over three decades has shown that groups' decisions are usually among the worst decisions. In fact, groups, instead of offsetting each others' biases, end up amplifying the problem! Groups tend to reduce the variance of opinions and for that reason members have more confidence in their decisions (**social proof**). Members enjoy enhanced competency and credibility in the eyes of their peers and do not like to provide information that runs contrary to the group's consensus. Think of analysts for whom producing forecasts too far away from their peers is very dangerous as they could lose their jobs if they are off. Hence, they tend to stay close to peers' forecasts very often. These forecasts are then passed on to investors or portfolio managers.

So, the lesson is that everyone involved in investing money—analysts, portfolio managers, and individual investors alike—suffers from behavioral biases that compromise one's ability to use information properly and to avoid the many traps that, to use James Montier's words, attract us onto the rocks.

5. Conclusion

This lesson was mainly about behavioral models and how they help financial advisors and portfolio managers to know their clients and to build a better relationship with them.

The next module will consider the problem of finding a better portfolio construction technique than mean-variance optimization and will look at the drawbacks of the optimal Sharpe portfolio. In addition, it will investigate what investment strategy is best aligned with a logarithmic utility function and what investment strategy results in maximum wealth.

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